SMART HELMET

An IoT-based Accident Detection and Emergency Alert System

Team 7
Section: M

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Abstract

Road accidents involving two-wheeler riders can result in serious injuries, especially when emergency assistance is delayed. This project presents a Smart Helmet designed to improve rider safety through automatic accident detection, wireless communication, emergency notification, and visual monitoring.

The proposed system uses an ESP32 as the main control unit and combines an MPU6050 accelerometer-gyroscope with an SW-420 vibration sensor to identify sudden impacts, abnormal acceleration, and unusual motion. A dual-validation mechanism is used to reduce false accident detection. When a possible accident is detected, the system provides an audible and visual warning and starts a short confirmation period. If the rider does not cancel the alert, the ESP32 sends an accident trigger to a paired smartphone through Bluetooth Low Energy (BLE). The smartphone can then obtain its GPS coordinates and send an emergency SMS or Telegram notification to predefined contacts.

An additional visual monitoring subsystem is incorporated using an ESP32-CAM and ESP32-CAM-MB. The ESP32-CAM can capture an image or short visual record when an accident event is confirmed, providing useful visual evidence of the incident. The ESP32-CAM-MB is used as the programming and convenient USB-to-serial interface board for the ESP32-CAM during development.

The overall system is designed to be compact, affordable, and suitable for practical prototype development. It demonstrates the integration of embedded systems, sensors, wireless communication, mobile automation, and IoT-based safety features.

Chapter 1

Introduction

Motorcycle and other two-wheeler riders are exposed to significant risks on the road. In many accidents, the injured rider may be unable to call family members or emergency services manually. Therefore, an automatic system capable of detecting an accident and initiating an emergency response can potentially reduce response time.

The proposed Smart Helmet combines a conventional protective helmet with embedded electronics. The system continuously monitors motion and vibration using an MPU6050 and an SW-420 vibration sensor. The ESP32 processes these sensor signals and determines whether an abnormal event should be treated as a possible accident.

The system also provides wireless connectivity. The ESP32 uses Bluetooth Low Energy (BLE) to communicate with a smartphone. After a confirmed accident, the smartphone can obtain location information and forward an emergency message to predefined contacts through SMS or Telegram.

To improve the information available after an incident, an ESP32-CAM is added to the system. It can capture visual information surrounding the helmet when an accident is confirmed. This feature can provide additional context about the incident and can be useful for later review.

The complete system therefore combines accident detection, user intervention, emergency communication, location sharing, and visual evidence collection in a single prototype.

Chapter 2

Project Objectives

The main objectives of the Smart Helmet are:

1. To detect possible motorcycle accidents automatically.
2. To monitor sudden acceleration, tilt, and vibration.
3. To reduce false accident alerts using dual-sensor validation.
4. To provide an emergency warning before sending an alert.
5. To allow the rider to cancel a false alarm manually.
6. To communicate with a smartphone using BLE.
7. To transmit accident information and location to predefined contacts.
8. To provide visual evidence using an ESP32-CAM.
9. To maintain a compact and low-cost embedded system design.
10. To demonstrate practical use of microcontrollers, sensors, communication modules, and IoT technologies.

Chapter 3

System Overview

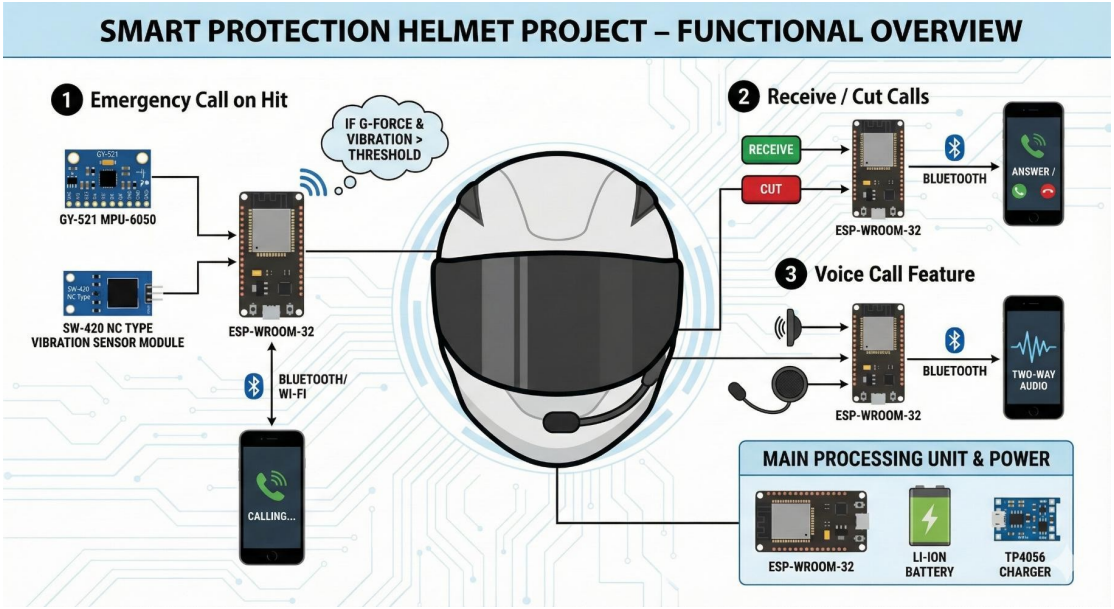


Figure 3.1: Overall functional overview of the Smart Helmet system

The Smart Helmet consists of several hardware and software modules. The ESP32 acts as the primary controller. The MPU6050 and SW-420 provide accident-related sensor information. A buzzer and LED provide local warning feedback. A manual cancel button gives the rider an opportunity to dismiss a false alarm.

The ESP32 communicates with a smartphone using BLE. The smartphone can perform GPS acquisition and long-distance communication through SMS or Telegram. The newly added ESP32-CAM subsystem provides image capture capability.

3.1 Overall System Flow

The basic operating sequence is:

1. The system starts and initializes all sensors and communication modules.

2. The MPU6050 continuously measures acceleration and orientation.
3. The SW-420 monitors sudden mechanical vibration or impact.
4. The ESP32 evaluates the sensor readings.
5. If an abnormal event is detected, the system enters a confirmation phase.
6. The buzzer and LED provide an immediate warning.
7. The rider can press the cancel button if no accident has occurred.
8. If the rider does not cancel the alert within the confirmation period, the accident is confirmed.
9. The ESP32 sends an accident trigger through BLE.
10. The paired smartphone obtains location information and sends an emergency message.
11. The ESP32-CAM captures visual information related to the accident event.

Chapter 4

Hardware Components

Component	Purpose
ESP32	Main controller, sensor processing, BLE communication
MPU6050	Measures acceleration, angular velocity, tilt, and motion
SW-420	Detects sudden vibration and mechanical shock
ESP32-CAM	Captures images/visual information during an accident event
ESP32-CAM-MB	USB programming and convenient serial interface for ESP32-CAM
GSM/Smartphone communication	Emergency communication mechanism
GPS/Smartphone GPS	Provides location information
Piezo Buzzer	Audible warning during accident confirmation
LED	Visual warning/status indication
Push Button	Manual false-alarm cancellation
3.7V Li-ion Battery	Portable power source
TP4056	Battery charging and protection module
MT3608	Voltage boost/regulation where required
Jumper Wires	Electrical connections
Foam Padding/Insulation	Mechanical protection and safe helmet integration

Chapter 5

ESP32 Main Controller



Figure 5.1: ESP32 development board used as the main controller

5.1 Purpose

The ESP32 is the central processing unit of the Smart Helmet. It receives data from the accident detection sensors, applies the programmed detection logic, controls the warning devices, and establishes BLE communication with the smartphone.

5.2 Key Features

- Dual-core microcontroller.
- Built-in Wi-Fi and Bluetooth/BLE support.
- Multiple GPIO pins for sensor and actuator interfacing.
- Low-power operation suitable for portable applications.
- Suitable processing capability for real-time sensor monitoring.

The ESP32 is selected because it combines processing capability and wireless connectivity in a compact and affordable platform.

Chapter 6

Accident Detection Subsystem

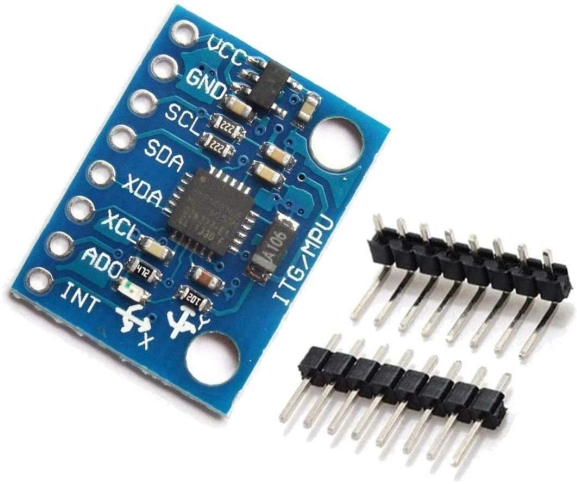


Figure 6.1: MPU6050 accelerometer and gyroscope module

6.1 MPU6050

The MPU6050 provides six degrees of freedom through a 3-axis accelerometer and a 3-axis gyroscope. It is used to monitor acceleration, angular motion, orientation, tilt, and sudden changes in movement.

A large acceleration spike can indicate a collision or fall. Similarly, a significant orientation change can provide additional evidence that the helmet has experienced an abnormal event.

6.2 SW-420 Vibration Sensor

The SW-420 is used as a secondary impact sensor. It responds to vibration and sudden mechanical shock. Its digital output makes it simple to interface with the ESP32.

The sensor is used together with the MPU6050 rather than as the only accident detector. This helps reduce the possibility that a minor movement will immediately generate an

emergency alert.

6.3 Dual-Validation Logic

The proposed detection mechanism uses multiple sensor conditions. A simplified version of the logic is:

$$AccidentCandidate = (GForce > 3G) \land (Vibration = HIGH) \tag{6.1}$$

where the approximate acceleration threshold is:

$$3G \approx 29.4\, m/s^2 \tag{6.2}$$

The exact threshold can be calibrated experimentally because different helmets, roads, speeds, and mounting positions can produce different sensor readings.

Chapter 7

Emergency Alert Mechanism

7.1 Confirmation Window

Immediately after a potential accident is detected, the system does not send the emergency message instantly. Instead, it activates a buzzer and LED and starts a short confirmation window.

The proposed confirmation period is 10 seconds.

7.2 False Alarm Cancellation

A manual button is provided for the rider. If the rider is safe and the detection was caused by a false trigger, pressing the button during the confirmation window cancels the emergency process.

This mechanism is important because vibration and sudden movement can sometimes occur without a real accident.

7.3 BLE Communication

If the confirmation timer expires without cancellation, the ESP32 sends an accident notification to the paired smartphone using BLE.

A sample trigger string is:

```
ACCIDENT_DETECTED
```

The smartphone automation system can use this trigger to start the emergency response process.

Chapter 8

Smartphone and Location-Based SOS

The smartphone is used to perform functions that would otherwise require additional hardware and power consumption on the helmet.

The smartphone can:

- Receive the BLE accident trigger.
- Obtain the current GPS coordinates.
- Generate a Google Maps location link.
- Send an emergency SMS.
- Send a Telegram notification.

An automation application such as MacroDroid can be configured to listen for the BLE trigger and execute the required emergency actions.

The general communication chain is:

$$\text{Sensors} \rightarrow \text{ESP32} \rightarrow \text{BLE} \rightarrow \text{Smartphone} \rightarrow \text{GPS} \rightarrow \text{SMS/Telegram}$$

Chapter 9

ESP32-CAM Visual Monitoring Feature

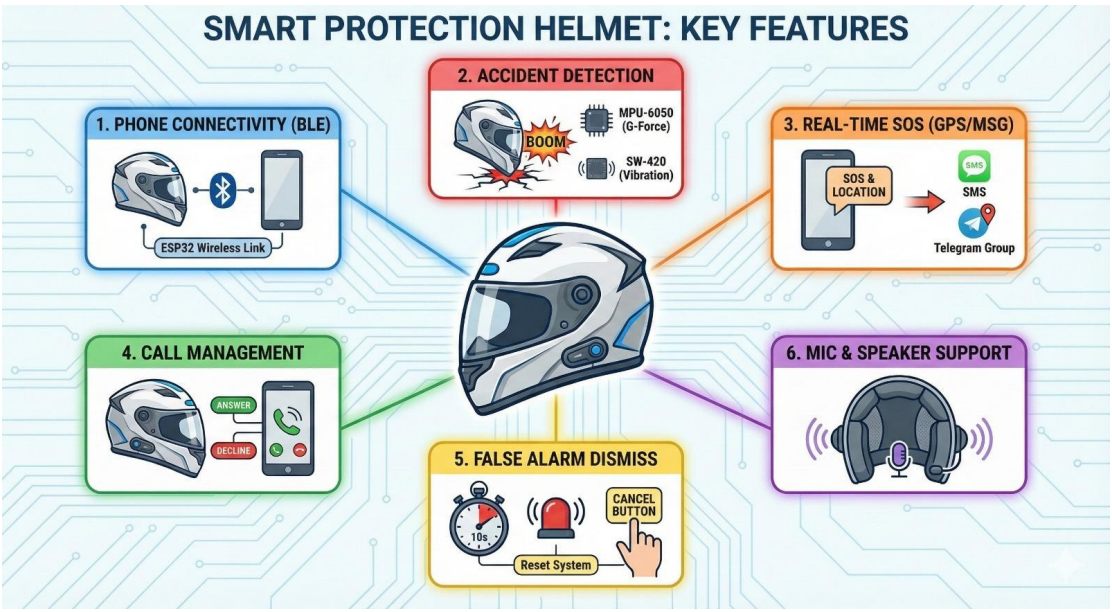


Figure 9.1: Key features of the proposed Smart Protection Helmet

9.1 Purpose of ESP32-CAM

An ESP32-CAM is added as an additional safety and monitoring feature. While the main ESP32 is responsible for sensor processing and emergency communication, the ESP32-CAM provides visual capture capability.

The purpose of this module is to record or capture visual information around the time of an accident. The captured image can provide useful contextual information such as the surrounding road condition, nearby objects, or the general scene at the time of the incident.

9.2 Accident-Triggered Image Capture

The ESP32-CAM can be configured so that an accident event causes the camera subsystem to capture an image.

A simplified sequence is:

1. MPU6050 and SW-420 detect an abnormal event.
2. ESP32 performs the accident validation.
3. The 10-second confirmation period begins.
4. If the rider does not cancel the alert, the accident is confirmed.
5. The ESP32 sends a trigger to the camera subsystem.
6. ESP32-CAM captures an image.
7. The image can be stored locally or transferred through an available network connection, depending on the final implementation.

The exact storage or transmission method depends on the final hardware and software implementation.

9.3 ESP32-CAM-MB

The ESP32-CAM-MB is used as a convenient programming and serial interface board for the ESP32-CAM.

Its main development purposes include:

- Easier USB connection during programming.
- Serial communication with the ESP32-CAM.
- Simplified firmware uploading during development.
- Convenient debugging and testing.

The ESP32-CAM-MB is therefore mainly a development/interface accessory, while the ESP32-CAM itself performs the actual image capture.

9.4 Role of the Camera Feature in the Overall System

The camera does not replace the accident detection sensors. Instead, it complements them.

Table 9.1: Role of the Main Safety Modules

Module	**Primary Role**
MPU6050	Motion, acceleration, tilt and impact-related measurement
SW-420	Vibration and mechanical shock detection
ESP32	Sensor processing, decision making and BLE communication
ESP32-CAM	Visual image capture
ESP32-CAM-MB	USB programming/serial interface for ESP32-CAM
Smartphone	GPS acquisition and emergency communication
Buzzer/LED	Local accident warning
Cancel Button	False alarm dismissal

Chapter 10

Hands-Free Communication

The proposed helmet can support hands-free communication using Bluetooth connectivity. A microphone and internal speaker can be integrated into the helmet padding.

The purpose is to allow the rider to receive or manage calls while minimizing the need to physically handle a smartphone during riding.

The communication feature can be implemented using a suitable Bluetooth audio module or an appropriate ESP32-based audio interface, depending on the final prototype design.

Chapter 11

Power Management



Figure 11.1: TP4056 lithium-ion battery charging module

The system is intended to operate from a rechargeable 3.7V Li-ion battery.

The TP4056 module is used for battery charging and protection. Where a stable higher voltage is required, an MT3608 boost converter can be used.

The simplified power structure is:

Li-ion Battery → TP4056 → Voltage Regulation → ESP32/Sensors/Actuators

Power consumption is an important consideration because the system is intended to be mounted on a helmet. The design therefore attempts to use wireless communication and processing efficiently.

Chapter 12

Software Architecture

The software can be divided into the following modules:

1. Sensor initialization.
2. Continuous motion and vibration monitoring.
3. Accident candidate detection.
4. Confirmation timer.
5. Manual cancellation handling.
6. BLE communication.
7. Smartphone automation.
8. Camera trigger and image capture.

12.1 Main Algorithm

1. Initialize ESP32, MPU6050, SW-420, buzzer, LED, button and BLE.
2. Initialize the ESP32-CAM subsystem.
3. Continuously read MPU6050 acceleration and orientation.
4. Read the SW-420 vibration status.
5. Check whether the predefined accident conditions are satisfied.
6. If not satisfied, continue monitoring.
7. If satisfied, activate buzzer and LED.

8. Start the 10-second confirmation timer.
9. Check the cancel button during the timer.
10. If the button is pressed, cancel the alert and return to monitoring.
11. Otherwise, confirm the accident.
12. Send the BLE accident trigger to the smartphone.
13. Trigger the ESP32-CAM to capture visual information.
14. Allow the smartphone automation system to obtain GPS coordinates.
15. Send an emergency SMS or Telegram notification.
16. Return to the appropriate system state after the emergency sequence.

Chapter 13

System Flowchart Description

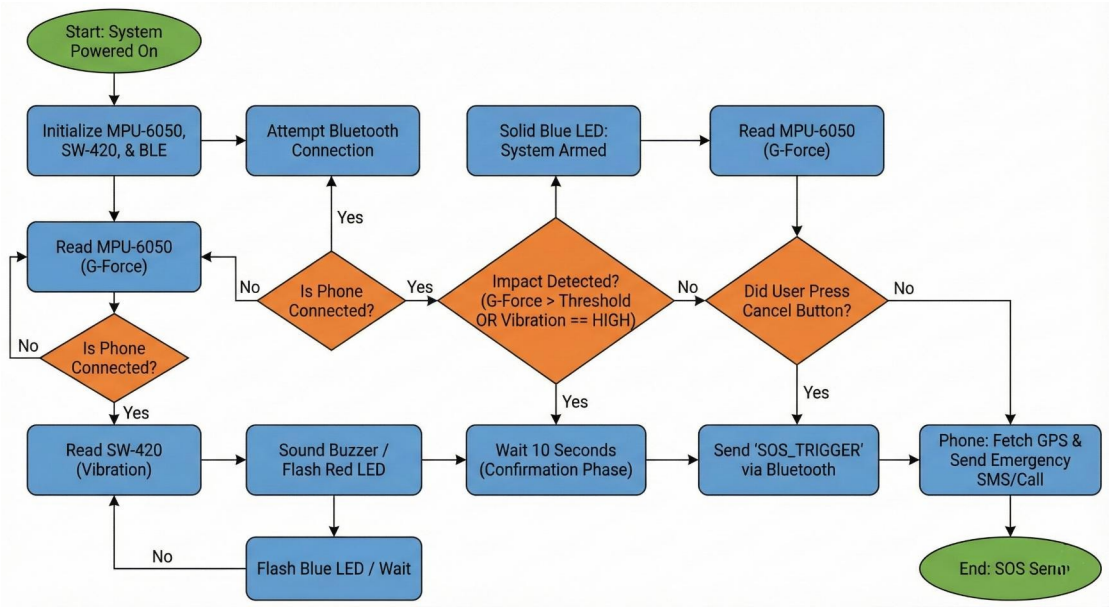
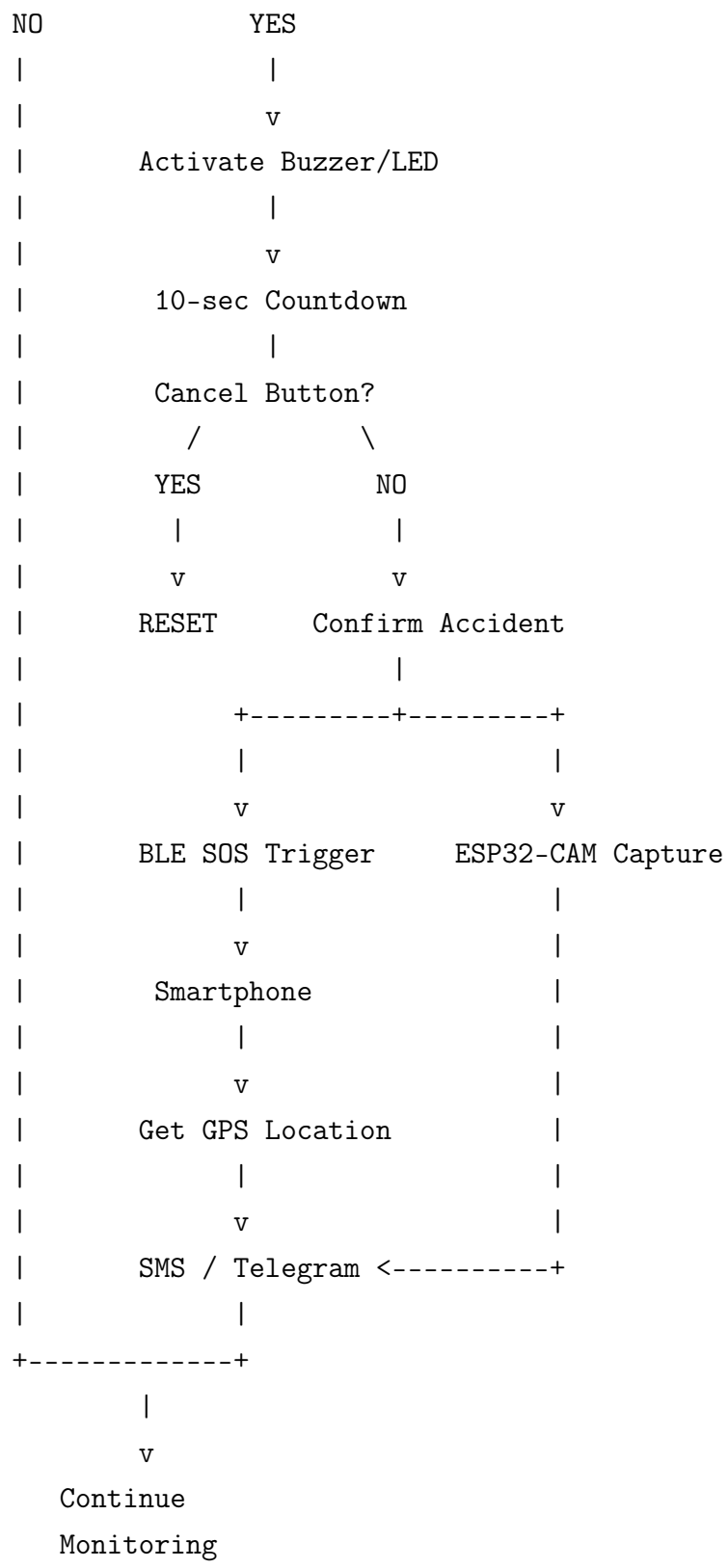


Figure 13.1: Smart Helmet accident detection and emergency response flowchart

The system flow can be represented as:

```
                START
                  |
                  v
          Initialize System
                  |
                  v
       Read MPU6050 + SW-420
                  |
                  v
        Accident Condition?
            /           \
```


NO
YES
Activate Buzzer/LED
10-sec Countdown
Cancel Button?
YES
NO
RESET
Confirm Accident
BLE SOS Trigger
ESP32-CAM Capture
Smartphone
Get GPS Location
SMS / Telegram
Continue
Monitoring

Chapter 14

Testing and Evaluation

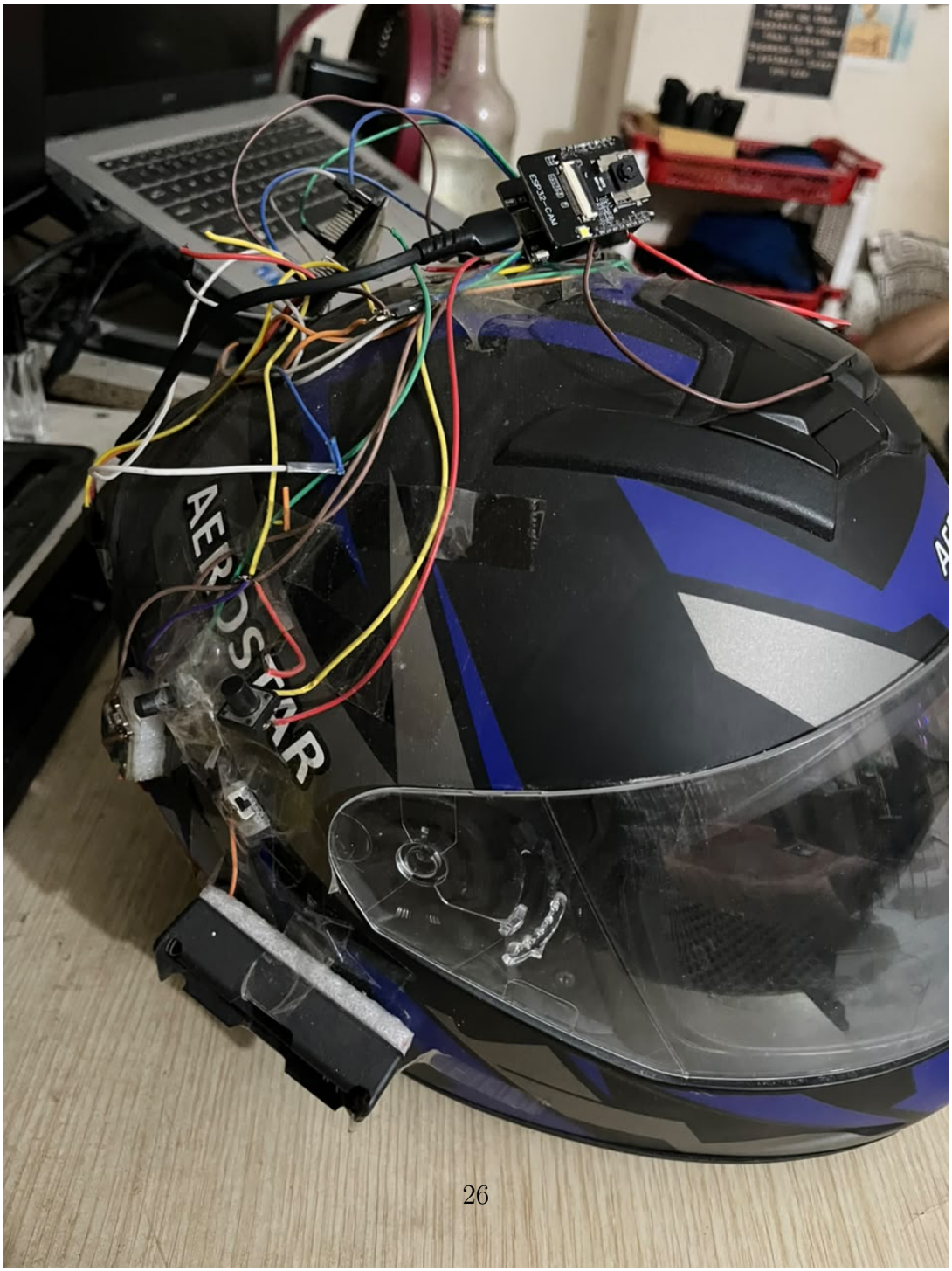


Figure 14.1: Developed Smart Helmet prototype with integrated electronics and ESP32-

The prototype should be evaluated under controlled conditions before real-world use.

14.1 Sensor Testing

The MPU6050 should be tested for:

- Sudden acceleration.
- Helmet tilt.
- Normal riding movement.
- Controlled impact simulation.

The SW-420 should be tested for:

- Low vibration.
- Strong vibration.
- Controlled impact.
- False-trigger conditions.

14.2 Emergency System Testing

The following scenarios should be tested:

1. Normal riding with no alert.
2. Minor movement without emergency notification.
3. Simulated accident with button cancellation.
4. Simulated accident without cancellation.
5. Successful BLE communication.
6. Successful GPS acquisition.
7. Successful SMS/Telegram notification.
8. Successful ESP32-CAM image capture.

14.3 Camera Testing

The ESP32-CAM should be evaluated for:

- Successful initialization.
- Image capture after accident trigger.
- Camera response time.
- Image quality under daylight and low-light conditions.
- Reliability of the trigger mechanism.
- Storage or transmission reliability.

Chapter 15

Advantages

The proposed system has several advantages:

- Automatic accident detection.
- Reduced dependence on manual emergency calls.
- Dual-sensor validation for improved reliability.
- Manual cancellation of false alarms.
- BLE-based smartphone connectivity.
- GPS-based emergency location sharing.
- SMS and Telegram emergency notification capability.
- Visual evidence through ESP32-CAM.
- Compact and relatively low-cost hardware.
- Practical hands-on implementation of embedded and IoT concepts.

Chapter 16

Limitations

The prototype also has limitations.

- Sensor thresholds require calibration for different riding conditions.
- Smartphone availability is required for the proposed GPS and long-distance alert mechanism.
- BLE communication depends on the connection between helmet and smartphone.
- Camera operation can increase power consumption.
- Low-light conditions may reduce image quality.
- The prototype should not be considered a certified medical or emergency-response device.
- Real-world reliability requires extensive testing under safe and controlled conditions.

Chapter 17

Future Improvements

Possible future improvements include:

1. Adding a dedicated GPS module for independent location tracking.
2. Adding cellular communication directly to the helmet.
3. Improving the accident detection algorithm using machine learning.
4. Adding cloud-based incident storage.
5. Uploading camera images automatically after an accident.
6. Adding a larger emergency contact management system.
7. Optimizing power consumption for longer battery life.
8. Adding waterproof and weather-resistant packaging.
9. Improving camera placement and image quality.
10. Developing a dedicated mobile application instead of relying on automation software.

Chapter 18

Conclusion

The Smart Helmet project demonstrates how microcontrollers, motion sensors, vibration sensors, wireless communication, mobile automation, and camera technology can be combined to improve rider safety.

The ESP32 serves as the main controller and processes data from the MPU6050 and SW-420 to identify possible accidents. A confirmation mechanism and manual cancellation button help reduce false emergency alerts. After a confirmed accident, BLE communication allows the system to notify a paired smartphone, which can obtain GPS coordinates and send emergency information through SMS or Telegram.

The addition of the ESP32-CAM extends the system beyond simple accident detection by providing visual information related to an incident. The ESP32-CAM-MB simplifies programming and serial communication during development. Together, these modules create a more informative and practical safety prototype.

Overall, the proposed Smart Helmet is a low-cost embedded and IoT-based safety system that demonstrates the practical application of microprocessors and microcontrollers in real-world problem solving.

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